

Western Mathematics Exams

School Name

Yearly Examination

2016

Year 10

Mathematics Course

Solutions

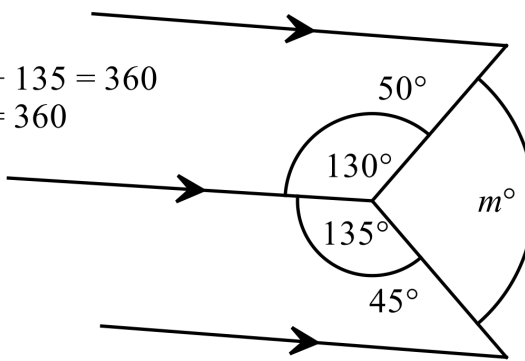
Year 10

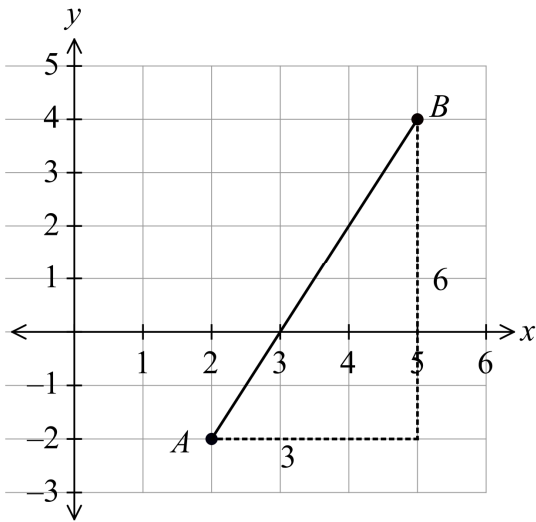
WME Solutions Mathematics Yearly 2016

Non Calculator

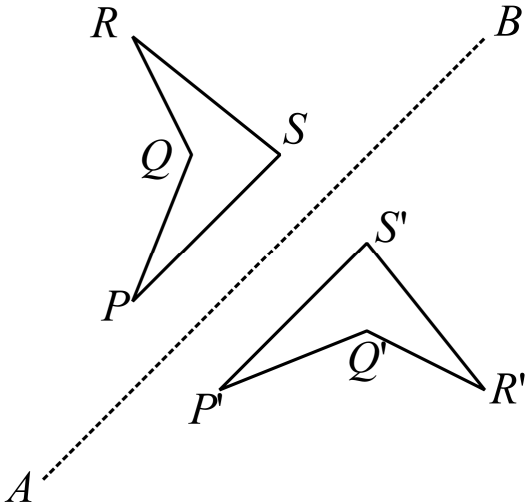
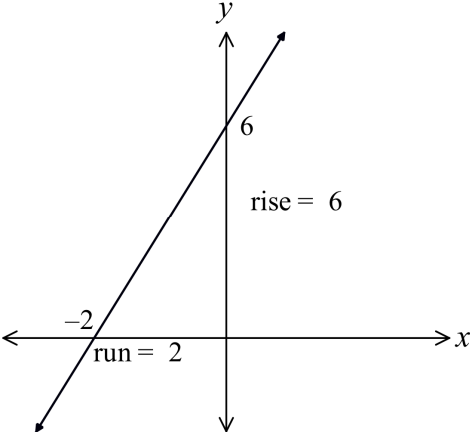
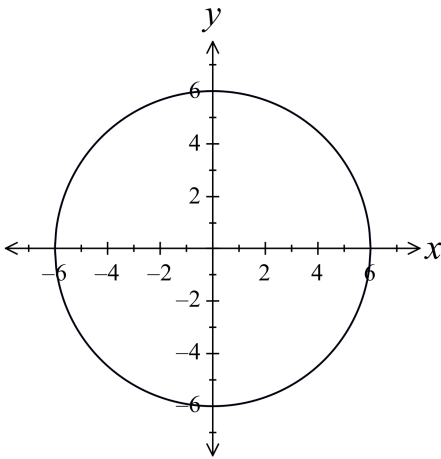
Section 1 Short Answer Section

ANSWERS

No.	WORKING	ANSWER
1.	$\begin{aligned}\text{Percentage children} &= \frac{28}{35} \times 100\% \\ &= \frac{4}{5} \times 100\% \\ &= 80\%\end{aligned}$	80%
2.	$\begin{aligned}\frac{7}{8} \text{ of } 72 &= \frac{7}{8} \times 72, \\ &= 63\end{aligned}$	63
3.	$\begin{aligned}\frac{-42 + 18}{-3} &= \frac{-24}{-3} \\ &= 8\end{aligned}$	8
4.	$\begin{aligned}m + 130 + 135 &= 360 \\ m + 265 &= 360 \\ m &= 95\end{aligned}$ 	$m = 95$
5.	$\begin{aligned}x + 2x + 3x + 4x &= 360^\circ \text{ (angle sum quadrilateral)} \\ 10x &= 360 \\ x &= 36\end{aligned}$	$x = 36$
6.	Glasgow is 9 hours behind, so Sydney is 9 hours ahead of 1:45 pm, which is 10:45 pm on Saturday 12 th November.	10:45 pm on Saturday 12 th Nov

7.	$\text{Area} = \frac{h}{2}(a + b)$ $540 = \frac{h}{2}(24 + 36)$ $540 = \frac{h}{2}(60)$ $30h = 540$ $h = \frac{540}{30} = 18 \text{ cm}$	18 cm
8.	Area of top of slab = 5 squares each $3 \text{ m} \times 3 \text{ m}$ $= 5 \times 9$ $= 45 \text{ m}^2$ Volume of Slab = 45×0.2 $= 9.0 \text{ m}^3$	9 m^2
9.	$T = \frac{mv^2}{L}$ $= \frac{100 \times 3^2}{6}$ $= \frac{\cancel{100} 50 \times \cancel{3} 3}{\cancel{6} 2} \quad \text{OR} = \frac{900}{6}$ $= 150$	150
10.	$4ab - 3a(2a - 4b) = 4ab - 3a(2a - 4b)$ $= 4ab - 6a^2 + 12ab$ $= 16ab - 6a^2$	$16ab - 6a^2$
11.	Gradient = $\frac{\text{rise}}{\text{run}}$ $= \frac{6}{3}$ $= 2$ 	2
12.	$27^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{27}} = \frac{1}{3}$	$\frac{1}{3}$

13.	$3d - 12 = 18 - 2d$ $5d - 12 = 18$ $5d = 30$ $d = \frac{30}{5} = 6$	$d = 6$
14.	Total number of Photos = $12 + 18 + 15 + 16 + 9 + 10 = 80$ Percentage from Bea = $\frac{16}{80} \times \frac{100}{1} = 4 \times 5 = 20\%$	20%
15.	$\text{Mean} = \frac{80}{6} = 13\frac{2}{6} = 13\frac{1}{3}$	$13\frac{1}{3}$
16.	$2 \text{ hours and } 20 \text{ min} : 1 \text{ hour and } 45 \text{ min} = 140 \text{ min} : 105 \text{ min}$ $= \frac{140}{5} : \frac{105}{5}$ $= 28 : 21$ $= 4 : 3$	4 : 3
17.	Midpoint of interval joining $P(-6, 10)$ and $Q(4, 5)$ $MP = \left(\frac{-6 + 4}{2}, \frac{10 + 5}{2} \right)$ $= \left(\frac{-2}{2}, \frac{15}{2} \right)$ $= \left(-1, 7\frac{1}{2} \right)$	$\left(-1, 7\frac{1}{2} \right)$
18.	$\cos A = \frac{A}{H} = \frac{15}{17}$	$\frac{15}{17}$
19.	$\text{Surface Area} = 22 \times 10 \times 2 + 22 \times 20 \times 2 + 10 \times 20$ $= 440 + 880 + 200$ $= 1\,520 \text{ cm}^2$	$1\,520 \text{ cm}^2$
20.	$\text{Volume} = \frac{1}{3} \times A \times h$ $= \frac{1}{3} \times 3 \times 3 \times 1.2$ $= 3.6 \text{ m}^3$	3.6 m^3

21.		The image $P'Q'R'S'$ is drawn accurately.
22.	$P(\text{divisible by } 3) = \frac{3}{10}$	$\frac{3}{10}$
23.	Gradient $m = \frac{\text{rise}}{\text{run}}$ $= \frac{6}{2} = 3$ y intercept = 6 Equation $y = mx + b$ $y = 3x + 6$ 	$y = 3x + 6$
24.	$x^2 + y^2 = r^2$ $x^2 + y^2 = 6^2$ $r = 6$ Centre is the origin.	

25.

$$\frac{BX}{DX} = \frac{AX}{CX}$$

$$\frac{8+12}{12} = \frac{x+18}{20}$$

$$\frac{20}{12} = \frac{x+18}{18}$$

$$x+18 = 18 \times \frac{20}{12}$$

$$x+18 = 3\cancel{(18)} \times \frac{10\cancel{(20)}}{\cancel{12}}$$

$$x+18 = 30$$

$$x = 30 - 18 = 12$$

$$x = 12$$

WME Solutions
Mathematics
Yearly 2016

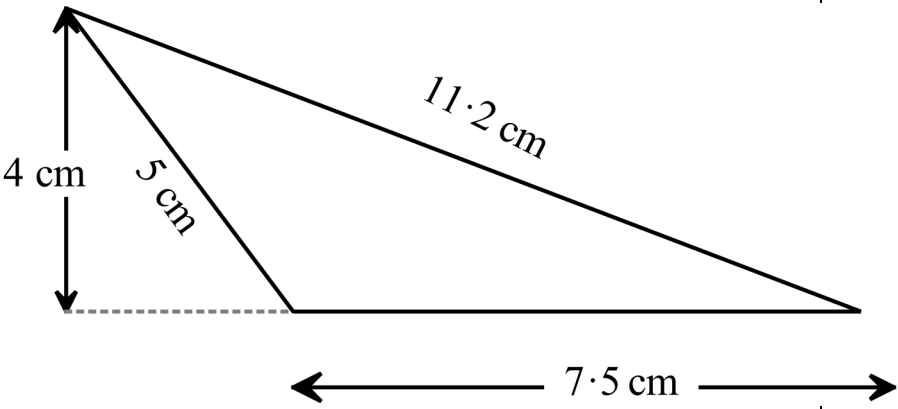
Calculator Allowed

Year 10

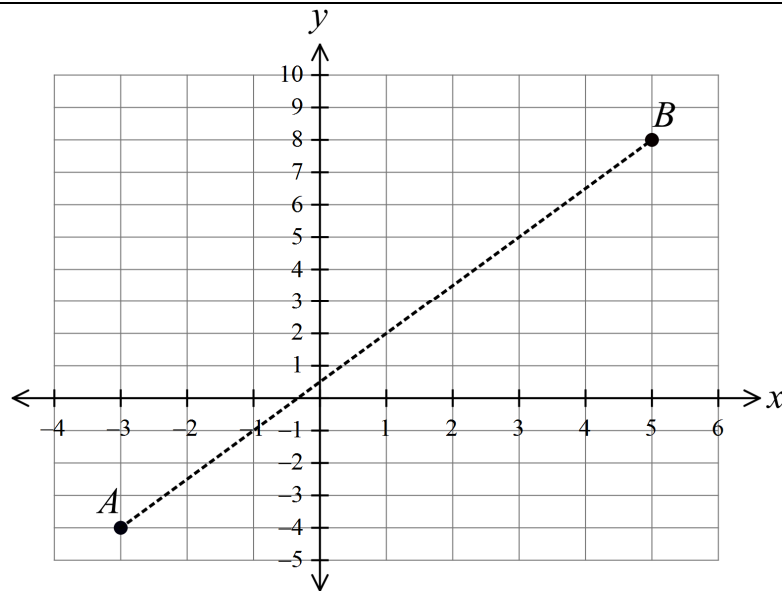
Section 2 Part A Multiple Choice Section

ANSWERS

No.	WORKING	ANSWER															
26.	$\frac{3}{8} = \frac{6}{16} = \frac{9}{24} = \frac{\mathbf{12}}{\mathbf{32}} = \frac{15}{40}$	D															
27.	$\begin{array}{r} 0.3 \\ 8 \overline{)2.4} \end{array}$ Amount in each glass is 0.3 litres	B															
28.	Amount after tax = $0.7 \times 112\,320$ = \$78 624 Weekly pay = $\$78\,624 \div 52$ = \$1 512	B															
29.	Profit = $264 - 160$ = \$104 Percentage profit = $\frac{104}{160} \times 100\%$ = 65%	C															
30.	<table><tr><td></td><td>Kite</td><td>Parallelogram</td><td>Rectangle</td><td>Square</td></tr><tr><td>The diagonals bisect one another.</td><td>N</td><td>Y</td><td>Y</td><td>Y</td></tr><tr><td>The diagonals are unequal.</td><td>Y</td><td>Y</td><td>N</td><td>N</td></tr></table>		Kite	Parallelogram	Rectangle	Square	The diagonals bisect one another.	N	Y	Y	Y	The diagonals are unequal.	Y	Y	N	N	B
	Kite	Parallelogram	Rectangle	Square													
The diagonals bisect one another.	N	Y	Y	Y													
The diagonals are unequal.	Y	Y	N	N													
31.	Each side measures 2 metres – 200 cm. Volume = $200^3 = 8\,000\,000\text{ cm}^3$ Capacity = $8\,000\,000\text{ mL}$ = $\frac{8000000}{1000} = 8000\text{ L}$	C															

32.	 Measurements could be taken in different directions to those shown, as long as the base and height are at right angles to one another. $\begin{aligned}\text{Area} &= \frac{1}{2} b h \\ &= \frac{1}{2} \times 7.5 \times 4 \\ &= 15 \text{ cm}^2\end{aligned}$	B
33.	$\begin{aligned}2az^2 - 15a^2 - 6az^2 + a^2 &= 2az^2 - 6az^2 - 15a^2 + a^2 \\ &= -4az^2 - 14a^2\end{aligned}$	D
34.	In $\triangle ABZ$ and $\triangle EDZ$ $\angle A = \angle E = 90^\circ$ $AZ = EZ$ $BZ = DZ$ $\therefore \triangle ABZ \equiv \triangle EDZ$ (RHS)	A
35.	$\begin{aligned}-4a(3ab - 5bc) &= -4a \times 3ab - 4a \times (-5bc) \\ &= -12a^2b + 20abc\end{aligned}$	A
36.	$\begin{aligned}30\% \text{ of total angle} &= 30\% \text{ of } 360^\circ \\ &= \frac{30}{100} \times 360 = 108^\circ\end{aligned}$ Fishing has an angle of 108°	A

37.



$$\begin{aligned}
 AB^2 &= 8^2 + 12^2 \\
 &= 64 + 144 \\
 &= 208 \\
 AB &= \sqrt{208}
 \end{aligned}$$

D

38.

$$\begin{aligned}
 \left(\frac{48x^{11}}{3x^3} \right)^{\frac{1}{2}} &= \left(16x^8 \right)^{\frac{1}{2}} \\
 &= 4x^4
 \end{aligned}$$

B

39.

$$\begin{aligned}
 2x &= \frac{2x - 10}{3} \\
 3 \times 2x &= \frac{3 \times (2x - 10)}{3} \\
 6x &= 2x - 10 \\
 4x &= -10 \\
 x &= -2.5
 \end{aligned}$$

B

40.

$$\begin{aligned}
 3 : 4.5 &= 30 : 45 & 3 : 4.8 &= 30 : 48 \\
 &= 2 : 3 & &= 5 : 8 \\
 3 : 5.4 &= 30 : 54 & 3 : 5.5 &= 30 : 55 \\
 &= \mathbf{5 : 9} & &= 6 : 11
 \end{aligned}$$

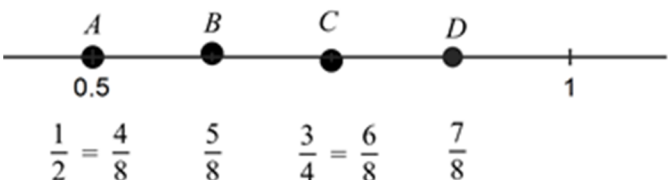
C

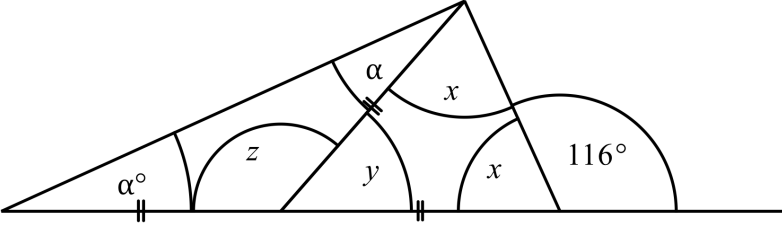
41.

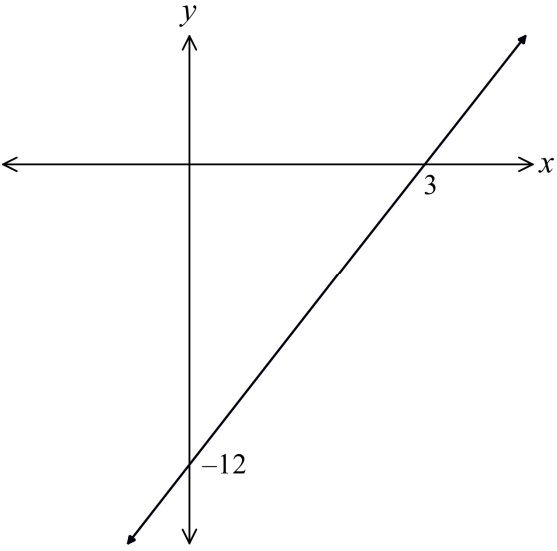
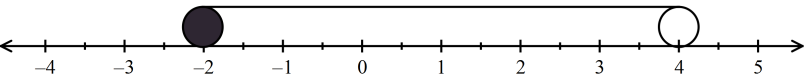
There are 20 with less than thirty spots.
 There are 25 in the survey altogether.
 Fraction with less than 30 spots = $\frac{20}{25} = \frac{4}{5}$

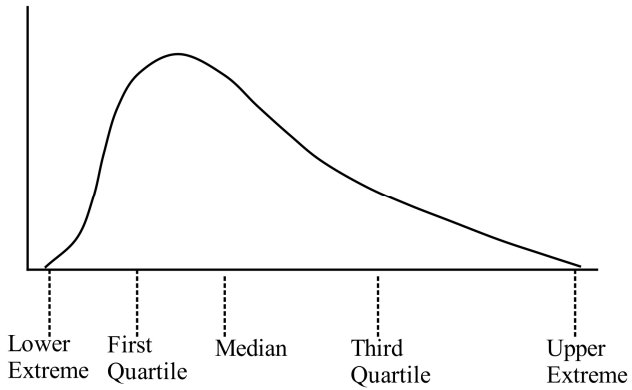
D

42.	Number of Spots Median = 27 (13th score with 12 above and 12 below) Lower Quartile = (middle of lower 12 scores so 6th and 7th) = 26 The median number of spots was 27 and the lower quartile was 26	D
43.	$\tan \alpha^\circ = \frac{24}{60} = \frac{2}{5} = 0.4$ $\alpha = \tan^{-1}(0.4)$ $= 21.80141$ $= 22 \text{ (nearest whole number)}$	A
44.	$\cos 35^\circ = \frac{MN}{2.45}$ $MN = 2.45 \times \cos 35^\circ$ $= 2.00692251$ $= 2.01 \text{ km}$	C
45.	$A = P(1 + r)^n$ $= 30000(1.06)^4$ $= \$37874.3088$ $= \$37\,874.31$	D
46.	Area for one tube = $2\pi r h$ $= 2 \times \pi \times 1.2 \times 30$ $= 226.194671$ Area of 2 500 tubes = $226.194671 \times 2\,500$ $= 565486.677$ $= 565\,000 \text{ cm}^2 \text{ (nearest thousand cm}^2\text{)}$	C

47.	$\begin{aligned}\text{Area of cross section} &= 15 \times 24 + \frac{1}{2} \times 24 \times 15 \\ &= 360 + 180 \\ &= 540 \text{ cm}^2 \\ \text{Volume of prism} &= 540 \times 40 \\ &= 21\,600 \text{ cm}^3\end{aligned}$	C
48.	There are 13 letters in OPPORTUNITIES. There are 2 O's and 1 U. $P(O \text{ or } U) = \frac{3}{13}$	C
49.	A pair of similar triangles are formed. $\frac{h}{5} = \frac{18}{3} = 6$ $h = 6 \times 5 = 30 \text{ m}$	D
50.	She was away from home for 11.5 hours (7 before midday and 4.5 after) She stopped for half an hour on the way and was at the appointment for 2 hours a total of 2.5 hours. Time driving = $11.5 - 2.5 = 9$ hours	A
51.	 $\frac{1}{2} = \frac{4}{8}$ $\frac{5}{8}$ $\frac{3}{4} = \frac{6}{8}$ $\frac{7}{8}$	B
52.	$\begin{aligned}I &= PRN \\ 672 &= 6400 \times 0.06 \times N \\ 672 &= 384 \times N \\ N &= \frac{672}{384} \\ &= 1.75 \text{ years} \\ &= 1 \text{ year and 9 months}\end{aligned}$	B

53.	$\angle COD = 90 - 58 = 32^\circ$ (adjacent angles in right $\angle$) $\angle COE = 32 + 45 = 77^\circ$ (adjacent $\angle$ s) $\angle GOF = \angle COE = 77^\circ$ (vert opp $\angle$ s) $\angle EOF = 180 - 77 = 103^\circ$ (angles on straight line) OTHER POSSIBLE REASONING	A
54.	 $x = 180 - 116 = 64^\circ$ (angles on st line) $2x + y = 180^\circ$ (angles in isosceles Δ) $128 + y = 180$ $y = 180 - 128 = 52$ $2\alpha = 52$ (exterior angle isos Δ) $\alpha = 26^\circ$ (or find z first $z = 180 - 52 = 128$) (or $z = 2x = 2 \times 64 = 128$) $2\alpha + 128 = 180$ $2\alpha = 52$ $\alpha = 26$	B
55.	$SV = IV(1 - r)^n$ $SV = 36000(1 - 0.12)^4$ $= 36000(0.88)^4$ $= \$21\,589.03$	B
56.	$3x - 5x^2 + 2x(3 - 4x) = 3x - 5x^2 + 6x - 8x^2$ $= 9x - 13x^2$	D
57.	$8ax^3 - 12ax^2 = 4ax^2(2x - 3)$ $= 2ax \times 2x(2x - 3)$ So $2ax$, $4ax^2$, and $(2x - 3)$ are factors and $8x$ is not.	A
58.	$\frac{3x}{4} \div \frac{6x}{5} = \frac{3x}{4} \times \frac{5}{6x}$ $= \frac{15x}{24x}$ $= \frac{5}{8}$	A

59.	Gradient $m = \frac{12}{3}$ $m = 4$ y intercept $b = -12$ Equation $y = 4x - 12$ 	D
60.	Percentage = $\frac{6.42 \times 10^{23}}{5.97 \times 10^{24}} \times 100$ $= 10.753768844221105527638190954774$ $= 10.8\%$ (correct to 3 sig fig)	A
61.	$-2 \leq x < 4$ -2 is included, so is shaded and 4 is not so is not shaded, and x lies between these two, so the line joins -2 and 4. 	B

62.	As the lower quartile, lower extreme and median are more closely groups than the upper quartile and upper extreme, the scores are more bunched toward the lower values, so the tail is toward the top, so it is positively skewed. 	C
63.	67 is the first quartile, so three-quarters of the scores will be equal or higher. $\frac{3}{4} \times 48 = 36$	D
64.	Probability from Birch Downs = $\frac{65}{250} = 0.26$ Probability not from Birch Downs = $1 - 0.26 = 0.74$	C
65.	Bearing of Daedalus = $270^\circ - 34^\circ = 236^\circ$	D
66.	$\cos 25^\circ = \frac{140}{KL}$ $KL \times \cos 25^\circ = 140$ $KL = \frac{140}{\cos 25^\circ}$ $= 154.47291$ $= 154 \text{ m (nearest m)}$	A
67.	$\angle STW = 180 - (24 + 107) = 180 - 131 = 49^\circ$ $\angle STU = 180 - (24 + 49) = 180 - 73 = 107^\circ$ $\therefore$ The angles in ΔSTU are $24^\circ, 49^\circ$ and 107° and The angles in ΔSTW are $24^\circ, 49^\circ$ and 107° $\therefore \Delta STW \parallel \Delta STU$	A

68.	$\text{Speed} = \frac{750}{3} \text{ m/min}$ $= 250 \text{ m/min}$ $= 15000 \text{ m/h}$ $= 15 \text{ km/h}$	D
69.	Interest rate = 7.2% pa = 0.6% per month Period = 4 years = 48 months $A = P(1 + r)^n$ $= 30000(1.0006)^{48}$ $= \$39978.30061$ $= \$39\,978.30$	C
70.	$y = 2x + 4$ has a gradient $m = 2$ the parallel line will have the same gradient It will be of the form $y = 2x + b$ And since passes through $(2, -6)$ $-6 = 2(2) + b$ $-6 = 4 + b$ $b = -10$ Equation $y = 2x - 10$	C
71.	$\frac{3^{-5} \times 9}{\sqrt{3} \div 27} = \frac{3^{-5} \times 3^2}{3^{\frac{1}{2}} \div 3^3}$ $= \frac{3^{-5+2}}{3^{\frac{1}{2}-3}}$ $= \frac{3^{-3}}{3^{-2\frac{1}{2}}}$ $= 3^{-3 - -2\frac{1}{2}}$ $= 3^{-\frac{1}{2}}$ $= \frac{1}{\sqrt{3}}$	B

72.	Number of mistakes for those 16 and under – 17, 18, 18, 14, 17, 15, 15, 11 $\text{Mean} = \frac{17 + 18 + 18 + 14 + 17 + 15 + 15 + 11}{8}$ $= \frac{125}{8}$ $= 15.625$	D
73.	The dots on the graph quite clearly follow a linear shape (straight line) so it is a strong relationship, and the number of mistakes decreases as the age increases so the relationship is negative. So there is a strong negative relationship between age and number mistakes.	A
74.	The graph has a y intercept of about 11, so only B and D have this. The graph is exponential so equation is B $y = 2^x + 10$ Or by testing a second point e.g. when $x = 6$ B. $y = 2^6 + 10 = 64 + 10 = 74$ D. $y = 6^2 + 11 = 36 + 11 = 47$ On Graph, when $x = 6, y = 74$ so correct graph is B.	B
75.	A parabola has a term in x^2 (or y^2) only, so correct equation is $y = x^2 + 10$	C

School Name
Year 10 Mathematics Examination
Solutions 2016
Multiple Choice Section Answer Sheet

Name _____ **Marking Sheet** _____

Teacher _____

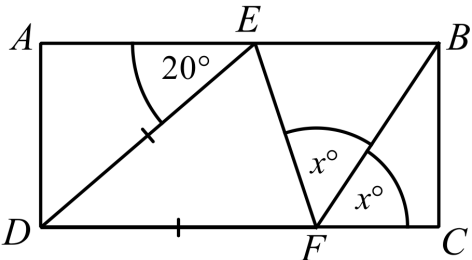
Completely fill the response oval representing the most correct answer.
Use a black or blue pen or 2B pencil.

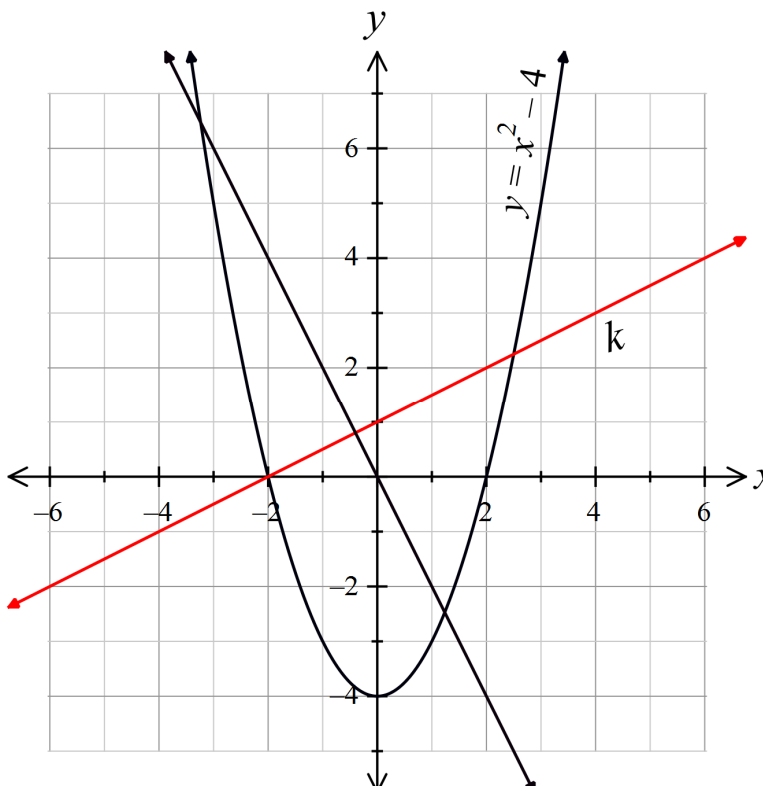
26. A ☐ B ☐ C ☐ D ☒
27. A ☐ B ☒ C ☐ D ☐
28. A ☐ B ☒ C ☐ D ☐
29. A ☐ B ☐ C ☒ D ☐
30. A ☐ B ☒ C ☐ D ☐
31. A ☐ B ☐ C ☒ D ☐
32. A ☐ B ☒ C ☐ D ☐
33. A ☐ B ☐ C ☐ D ☒
34. A ☒ B ☐ C ☐ D ☐
35. A ☒ B ☐ C ☐ D ☐
36. A ☒ B ☐ C ☐ D ☐
37. A ☐ B ☐ C ☐ D ☒
38. A ☐ B ☒ C ☐ D ☐
39. A ☐ B ☒ C ☐ D ☐
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45. A ☐ B ☐ C ☐ D ☒
46. A ☐ B ☐ C ☒ D ☐
47. A ☐ B ☐ C ☒ D ☐
48. A ☐ B ☐ C ☒ D ☐
49. A ☐ B ☐ C ☐ D ☒
50. A ☒ B ☐ C ☐ D ☐

51. A ☐ B ☒ C ☐ D ☐
52. A ☐ B ☒ C ☐ D ☐
53. A ☒ B ☐ C ☐ D ☐
54. A ☐ B ☒ C ☐ D ☐
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70. A ☐ B ☐ C ☒ D ☐
71. A ☐ B ☒ C ☐ D ☐
72. A ☐ B ☐ C ☐ D ☒
73. A ☒ B ☐ C ☐ D ☐
74. A ☐ B ☒ C ☐ D ☐
75. A ☐ B ☐ C ☒ D ☐

Year 10	WME Solutions Mathematics Yearly 2016		Calculator Allowed
Section 2 Part B Longer Answer Section			
ANSWERS			
			Marks
76.	a)	There are 21 scores, so 21 players	1 mark for correct answer
	b)	The sum of the scores = 1029 Mean = $\frac{1029}{21} = 49$	1 mark for correct answer
	c)	Median is the 11 th score (10 above and 10 below) Median = 48	1 mark for correct answer
	d)	Lower quartile = $\frac{39 + 40}{2} = 39.5$ (5 th and 6 th scores) Upper quartile = $\frac{56 + 58}{2} = 57$ (16 th and 17 th scores) Interquartile range = $57 - 39.5 = 17.5$	1 mark for correct answer
77.	a)	Volume = $A h$ $= \left(\frac{1}{2} \times 10 \times 12 \right) \times 16$ $= 960 \text{ cm}^3$	1 mark for correct answer
	b)	Find slant height l $l^2 = 5^2 + 12^2$ $= 25 + 144 = 169$ $l = \sqrt{169} = 13$ Surface Area = $\frac{1}{2} \times 10 \times 12 \times 2 + 16 \times 13 \times 2 + 10 \times 16$ $= 120 + 416 + 160$ $= 696 \text{ cm}^2$	2 marks for correct answer 1 mark for an answer calculated using the wrong slant height or which has another similar simple error.

78.	a)	$2(4p - 3) = 4p - 17$ $8p - 6 = 4p - 17$ $4p - 6 = -17$ $4p = -11$ $p = -2.75$	2 marks for correct answer found by any valid method including trial and error. 1 mark for an incorrect answer which has valid working toward answer with minor error in algebra or calculation
	b)	$(c + 4)(c - 2) = 0$ $c = -4 \text{ or } c = 2$	1 mark for correct answer
	c)	Sub $y = 4$ into $y = 3x - 5$ $4 = 3x - 5$ $3x = 9$ $x = 3$ Point (3, 4)	1 mark for correct answer
79.		$A = 59\,100$ Three years at 8% pa quarterly so $n = 3 \times 4 = 12$ $r = 0.08 \div 4 = 0.02$ $A = P(1 + r)^n$ $59\,100 = P(1.02)^{12}$ $59\,100 = 1.26824 \times P \text{ (rounded to 6 s. f.)}$ $P = \frac{59\,100}{1.26824}$ $P = 46\,599.95$ $P = 46\,600 \text{ (to nearest dollar)}$	2 marks for correct answer found by any valid method including trial and error using the formula. 1 mark for an incorrect answer which uses the compound interest formula but has an error in substitution, calculation or in changing to the quarterly rates.

80.		 $\angle EDF = \angle AED = 20^\circ$ (alt $\angle$ on $\parallel$ lines) $\angle DFE = \angle FED$ (isosceles Δ) $20^\circ + 2 \times \angle DFE = 180^\circ$ (angle sum Δ) $\angle DFE = \frac{180 - 20}{2} = 80^\circ$ $2x + 80 = 180$ ($\angle$ on straight line) $2x = 100$ $x = 50^\circ$	2 marks for correct answer with reasoning. 1 mark for an incorrect answer which has working which includes some correct and relevant reasoning
81.	a)	$\tan 40^\circ = \frac{101}{AD}$ $AD = \frac{101}{\tan 40^\circ}$ $= 120.3671128$ $= 120 \text{ m (nearest metre)}$	1 mark for correct answer
	b)	$\tan F = \frac{101}{160}$ $F = \tan^{-1} \left(\frac{101}{160} \right)$ $= 32.262169296$ $= 32^\circ \text{ (nearest degree)}$	1 mark for correct answer
	c)	$\angle EBA = 25^\circ$ $\sin \angle EBA = \frac{AE}{AB}$ $\sin 25^\circ = \frac{AE}{200}$ $AE = 200 \sin 25^\circ$ $= 84.52365$ $= 85 \text{ m (nearest metre)}$ Height of balloon above ground = $85 + 101 \text{ m}$ $= 186 \text{ m}$	2 marks for correct answer 1 mark for an answer with a minor error or which only finds the distance AE.
82.	a)	The intercepts are $x = -2$ and $x = 2$.	1 mark for correct answer

	b)	From the points $(-2, 0)$ and $(0, 1)$ Rise = 1 Run = 2 Gradient = $\frac{1}{2}$ Or using formula with two points $m = \frac{1-0}{0-(-2)} = \frac{1}{2}$	1 mark for correct answer
	c)	 Gradient = -2 (either from graph or $m_1 \cdot m_2 = -1$) y intercept = 0 Equation $y = -2x$	1 mark for correct line drawn on number plane 1 mark for correct equation
83.	a)	$P(\text{ Walking }) = 16\% + 28\% = 44\%$ Can be written as 0.44 or $\frac{11}{25}$	1 mark for correct answer
	b)	$P(\text{ Not Walking and Cycling }) = 100\% - 16\% = 84\%$ Can be written as 0.84 or $\frac{21}{25}$	1 mark for correct answer